

Symptomatic Patients With an Early Viable Intrauterine Pregnancy: hCG Curves Redefined

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OBJECTIVE: To analyze the change in serial human chorionic gonadotropin (hCG) levels in women symptomatic with pain or bleeding who presented with nondiagnostic ultrasonography but were ultimately confirmed to have a viable intrauterine pregnancy.

METHODS: The rise in serial hCG measures were modeled over time, with the start point defined in 2 ways: by last menstrual period and by date of presentation for care. Both semiparametric (spline) curves and linear random-effects models were explored. The slope and projected increase of hCG were calculated to define 99% of viable intrauterine pregnancies.

RESULTS: A total of 287 subjects met inclusion criteria and contributed 861 measurements of hCG. On average, these subjects contributed 3.00 observations and were followed up for 5.25 days. A linear increase in log hCG best described the pattern of rise. Curves derived from last menstrual period and day of presentation do not differ substantially. The median slope for a rise of hCG after 1 day was 1.50, (or a 50% increase); 2.24 after 2 days (or a 124% rise), and 5.00 after 4 days. The fastest rise was 1.81 at 1 day, 3.28 at 2 days, and 10.76 at 4 days. The slowest or minimal rise for a normal viable intrauterine pregnancy was 24% at 1 day and 53% at 2 days.

CONCLUSION: These data define the slowest rise in serial hCG values for a potentially viable gestation and will aid in distinguishing a viable early pregnancy from a miscarriage or ectopic pregnancy. The minimal rise in serial hCG values for women with a viable intrauterine pregnancy is "slower" than previously reported, suggesting that intervention to diagnosis and treat an abnormal gestation should be more conservative. (Obstet Gynecol 2004;104:50–5. © 2004 by The American College of Obstetricians and Gynecologists.)

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The discovery in the late 1920s that the urine of pregnant women had a pregnancy factor led to development of multiple bioassays, including the famous Aschheim-Zondek test.¹ In 1938, Gey et al² demonstrated the production of the molecule in trophoblast tissue culture. Since then, human chorionic gonadotropin (hCG) levels, and their change over time, have represented an invaluable aid in the evaluation of early gestation.

Human chorionic gonadotropin is detectable in the plasma of pregnant women 8 days after ovulation, which is likely the time of implantation.³ Levels increase during gestation and reach a peak of approximately 100,000 mIU/ml at 10 weeks, then decrease and remain stable at approximately 20,000 mIU/ml.

Multiple studies have evaluated the rise of hCG in normal and abnormal pregnancy. Currently, clinicians rely on a normal "doubling time," the time necessary for the hCG levels to double, to characterize a viable gestation when ultrasonography is not indicated or is nondiagnostic. Although there is consensus that the predictable rise in serial hCG values is distinct from the slow rise or plateau of a nonviable pregnancy such as a miscarriage or ectopic pregnancy,^{4–14} there has been disagreement on the rate of increase and type of curve that best describes the rise of a normal viable pregnancy.^{4–11} Pitfalls of previous studies have been the evaluation of relatively few women and study populations that may not be generalizable to women at risk for miscarriage or ectopic pregnancy. It is important that the often-quoted doubling time of an early gestation (66% rise in 2 days) is based on an 85% confidence interval of a study of only 20 women.⁴ If the "definition" of a nonviable pregnancy is too strict, viable gestations may be interrupted during the diagnosis and treatment of nonviable gestation such as a miscarriage or ectopic pregnancy.

The purpose of this study was to characterize the change in hCG levels in a large cohort of women with a spontaneously conceived pregnancy who are at risk for miscarriage or ectopic pregnancy because of symptoms of pain or bleeding in the first trimester, but who subse-



quently were confirmed to have a viable intrauterine pregnancy. Human chorionic gonadotropin curves established from this population should offer the most direct evidence to aid a physician in differentiating an early normally growing pregnancy from a nonviable pregnancy in these women at high risk for an abnormal pregnancy.

MATERIALS AND METHODS

This study was approved by the Institutional Review Board of the University of Pennsylvania (Risk Factors of Predictors of Ectopic Pregnancy, # 103700). The Hospital of the University of Pennsylvania uses a computerized database to track all pregnant women who presented to the Emergency Department at risk for an ectopic pregnancy or a miscarriage because of the symptoms of pelvic pain or vaginal bleeding in the first trimester. Data were abstracted from subjects who presented between January 1, 1990, and July 31, 1999. For this report, analysis was restricted to women for whom a diagnosis could not be made at their initial visit, and therefore were followed up with serial hCG measurements. Women were followed up until a definitive intrauterine pregnancy was diagnosed. An intrauterine pregnancy was defined as transvaginal ultrasound confirmation of gestational sac with a yolk sac or fetal pole.

To be included in this evaluation women had to have had at least 2 hCG determinations, at least 24 hours apart (but not more than 7 days), before the day of intrauterine pregnancy confirmation. Analysis was restricted to singleton pregnancies only. If women had 2 (or more) pregnancies recorded in the database during this time period, 1 was chosen at random and the other(s) excluded from analysis.

Serum hCG was determined at the Pepper Laboratory of the University of Pennsylvania, using the Abbott AxSYM (Abbott Laboratories, Abbott Park, IL) or DPC Immulite (Diagnostic Products Corporation, Los Angeles, CA) total β immunoassay. The interassay and intra-assay variation was less than 10%. Results are expressed as milli-International Units per milliliter, using the third international reference standard.

We evaluated the curve created by the serial hCG values. All analyses were performed on the natural log transformation of hCG values. This transformation was necessary to alleviate the skewness of the hCG distribution and reduce the influence of large values. The starting time of the curve was defined and evaluated in 2 ways. In one analysis the start time was a known date of last normal menstrual with a range of 22–95 days before presentation. However, in some cases the date of last menstrual period is unknown or inaccurate. To overcome this issue, an analysis was conducted using all

available subjects by considering the day of presentation for care as the start time. This aspect is important, because the shape of the curve (and especially whether the slope changes during observation) may depend on time since conception. If the curve for log hCG is non-linear, then the time definition will be critically important, because the change in serially measured values may depend not only on the time between measurements, but also on gestational age.

Using recently developed semiparametric statistical techniques, splines,^{14–17} the shape of the hCG curve was evaluated. Using these techniques enabled us to consider the appropriateness of a linear relationship through time or whether the function is more complicated (eg, increasing rapidly, falling off, or declining). The polynomial spline model is a modern statistical approach to fit smooth curves without being restrictive about the shape of the curves.¹⁸ In addition, a linear (parametric) model was also considered. Both the spline and linear models used random effects to account for the repeated hCG measurements contributed by each subject and are quite flexible even when the number and timing of observations are unbalanced.¹⁹ Both methods estimate a population average curve by aggregating the profile estimated for each subject individually. Curves generated using both linear and spline mixed-effect models are presented in Figure 1. To evaluate the shape of the curves, we overlaid the graphs of the estimated population profile along with 95% confidence intervals (CIs) for the average profile. Based upon the large amount overlap in the confidence intervals of these 2 curves, we determined that the linear fit was appropriate for describing these data (figure not shown).

From the linear random-effects model, population average values for increase or slope, standard errors, and upper and lower confidence bounds on the rate of increase in log hCG were estimated, which assumed that log hCG follows a normal distribution. This model was used to calculate expected values at important clinical time points after presentation for care. We present slopes for the increase in log hCG as well as the rates of increase for a 1-to-7-day range. In addition to the average values for increase, we present upper and lower confidence estimates (percentiles) on the rate of increase.

RESULTS

A total of 2,545 women were identified in the electronic database, with 147 women reporting more than 1 pregnancy during the study period. A subset of 1,543 subjects was identified in the data set for whom the diagnosis was not clinically apparent at presentation of pain or vaginal bleeding in conjunction with a positive pregnancy test.



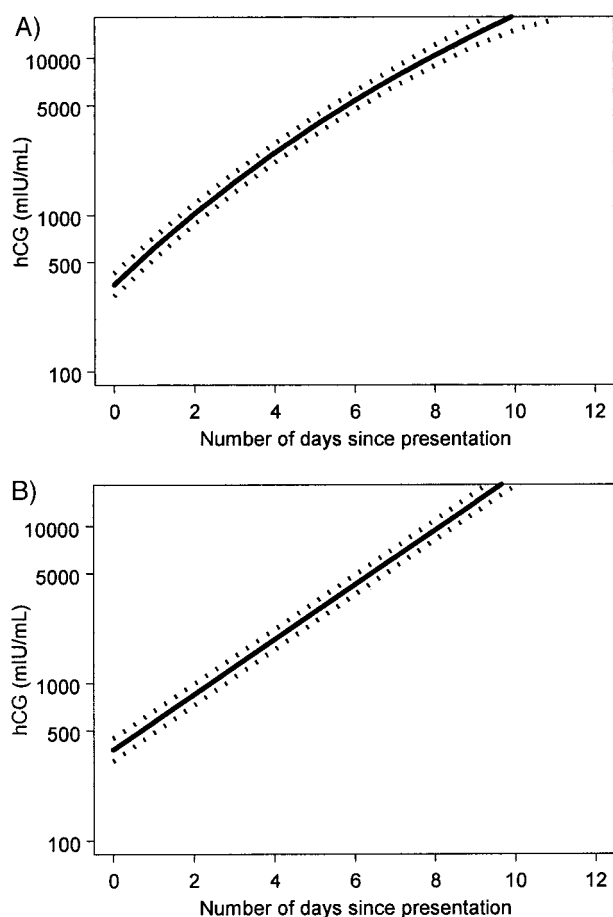


Fig. 1. Plot of serial values of log hCG concentrations and 95% confidence interval (*dotted lines*) for women (261 subjects with 774 observations) with a documented intrauterine pregnancy. **A.** Semiparametric spline fit. **B.** Linear (parametric) fit. Given very little difference between the 2 curves, a linear fit was chosen to best characterize the curve. Barnhart. *hCG Rise in Intrauterine Pregnancy. Obstet Gynecol* 2004.

Of those, 378 women were ultimately diagnosed with an intrauterine pregnancy with data fitting the protocol. Ninety-one of these women were excluded from analysis because the data were outside of the clinical area of interest (hCG > 5,000 mIU/mL at presentation, follow-up lasting greater than 3 weeks, or were ultimately diagnosed with a nonviable gestation after initial diagnosis of intrauterine pregnancy). Thus, the final sample included 287 subjects contributing 861 measurements of hCG.

The average age for subjects was 23.6 (range, 14.8–38.5) years. The majority of subjects were African American (87%). Average gravida was 2.45 (range, 1–12), with an average parity of 0.89 (range, 0–6). These subjects contributed on average 3.00 (range, 2–7) observations

each and were followed up an average of 5.25 (range, 1–14) days until the diagnosis of intrauterine pregnancy was confirmed. The average initial hCG value was 783 mIU/mL, with a standard deviation of 960. The median value was 388, with upper quartile of 1,118 and lower quartile of 119. The average estimated gestational age by reported last menstrual period (LMP) was 5 weeks and 3 days (range, 0–107 days).

The subset of women with known LMP (between 22 days and 96 days of presentation) included 261 women (91%) who contributed 774 hCG measurements. Figure 1 compares spline and linear mixed-effects model fits for the rate of change in log hCG. The time scale for both plots is measured in days since presentation for care. Given this scale, the assumption of linear increase in log hCG describes the data adequately. Figure 2 compares the curves derived using different start points (date of presentation or last menstrual period) after transformation back to the original scale of hCG (mIU/mL). When we considered the log-linear model fit to the subset of women who had known LMP along with date of presentation, the slopes for these 2 curves were nearly identical (estimated slope using LMP was 0.407 with 95% CI 0.395–0.419; estimated slope using presentation was 0.402 with 95% CI 0.389–0.415).

The final model was derived using all data, with date of presentation as the start point and assuming a linear model fit for log hCG. This curve, with confidence estimates for the potential slowest growth are illustrated in Figure 3. This figure is presented with hCG concentration as milli-International Units per milliliter (not log scale). The equation that describes the line is $\log(\text{hCG}) \text{ increase} = 0.402 \times \text{Day}$.

Using this final model, the expected rates of increase for hCG value are presented in Table 1. The results for the estimated rate of increase are virtually identical or perhaps marginally more conservative than that using LMP as start date. The table can be used clinically to compare the rate of change in serial hCG values or extrapolate expected serial hCG values to compare with initial values. For example, if subsequent measurements of hCG were taken 1 day after the first, one would anticipate that the level would have increased by a factor of 1.50, or a 50% increase, using the median estimate (50th percentile). If the measurements were taken 2 days apart, we would expect the value to have increased by factor of 2.24 or 124%. Values of hCG can rise as rapidly as 81% in 1 day, 228% in 2 days and nearly a 10-fold increase in 4 days. A total of 99% of viable pregnancies should have a 1-day increase of at least 1.24 (24%) or at least a 53% rise in 2 days (row 1 of Table 1, 1st percentile). In other words, a rise slower than this rate indicates the gestation was experiencing abnormal growth.



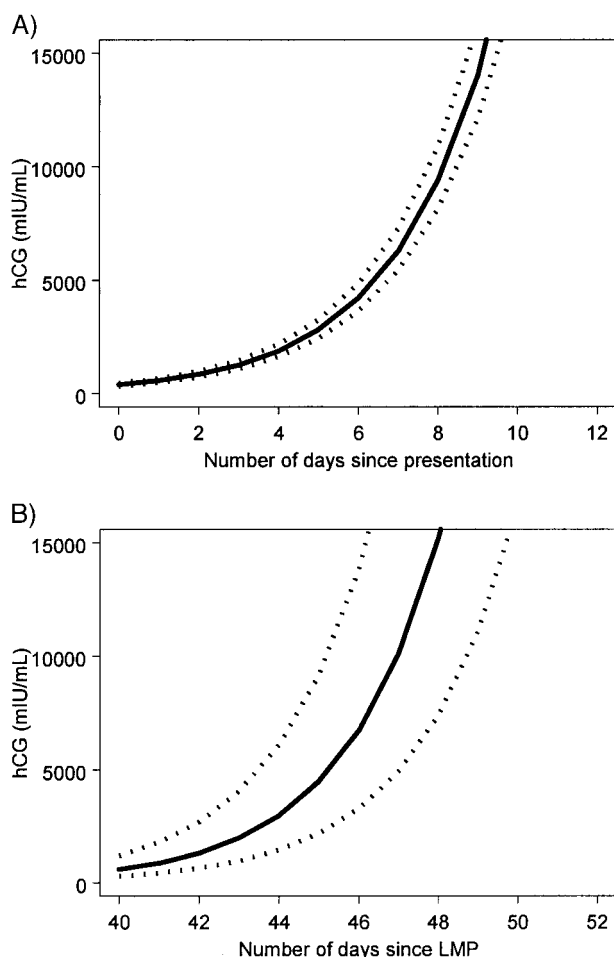


Fig. 2. Curve with 95% confidence interval (*dotted lines*) generated from serial hCG concentrations of women with a documented intrauterine pregnancy (261 subjects with 774 observations). **A.** Date of presentation of a woman with pain and/or bleeding is the starting time. **B.** Last menstrual period is the starting point.

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DISCUSSION

The goal of this study was to define the rate of increase of hCG values in a population of women in whom the diagnosis of an abnormal pregnancy is most difficult and problematic. Our analysis was restricted to women whose pregnancy was not initially identifiable with ultrasonography, whose initial hCG value was less 5,000 mIU/mL, and who presented with symptoms of pain or bleeding. We purposely chose a value for hCG slightly above the value of most discriminatory zones (the level of hCG that a viable intrauterine pregnancy should be when visualized using transvaginal ultrasonography) to allow for a margin of error in those clinical situations that may need a few extra days of surveillance before defini-

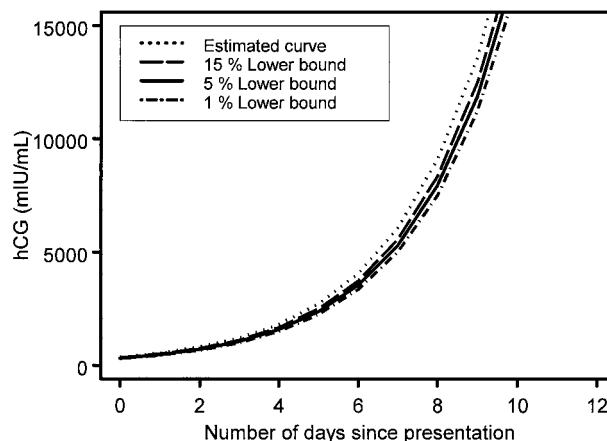


Fig. 3. Generated curve of serial hCG values of women (287 subjects with 861 observations) who present with pain and/or bleeding in the first trimester who ultimately were diagnosed with an intrauterine gestation using a linear model. Confidence intervals of the lower bound of increase are illustrated.

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tive diagnosis. All women were subsequently confirmed to have an intrauterine pregnancy.

Previous data evaluating the change in hCG for women undergoing fertility treatment or without symptoms of pain or bleeding in the first trimester may not be comparable with the clinically important population of symptomatic first trimester pregnancies. This is exemplified by our finding that the rate of hCG rise is slower than previously reported.⁴⁻⁹ This suggests that clinical practice may have been too aggressive in diagnosing women with a slow rise in serial hCG values as having a nonviable gestation. Potentially, this determination may result in the interruption of viable pregnancies in the attempt to diagnose or treat women for ectopic pregnancy or miscarriage.

Our investigation identifies a number of important aspects regarding the curve generated by serial hCG concentrations for these pregnancies. It has been suggested that the curve describing the rise in serial hCG in a normal pregnancy can be described as log linear,⁴⁻⁷ quadratic,⁸⁻⁹ or a combination of both, depending upon gestational age.¹⁰⁻¹¹ We note that in this restricted time frame, the curve generated for these serial values can best be represented by a log-linear model. This allows simpler extrapolation to future values. However, these data do not characterize the curve for gestations greater than 10 weeks from last menstrual period or those with an initial hCG of greater than 5,000 mIU/mL. It is possible that the curve describing the change in hCG for relatively advanced gestational age may be more complex.



Table 1. Predicted Increase in Log hCG Over Time

Percentile	Slope of hCG rise	Relative increase in hCG concentration from baseline				
		1 day later	2 days later	3 days later	4 days later	7 days later
1	1.24	1.24	1.53	1.88	2.33	4.38
5	1.31	1.31	1.71	2.23	2.91	6.49
10	1.35	1.35	1.81	2.44	3.28	8.00
15	1.37	1.37	1.89	2.59	3.56	9.21
50	1.50	1.50	2.24	3.35	5.00	16.73
85	1.63	1.63	2.65	4.32	7.04	30.39
90	1.66	1.66	2.76	4.59	7.63	35.00
95	1.71	1.71	2.93	5.02	8.60	43.15
99	1.81	1.81	3.28	5.94	10.76	63.88

Percentile of 50 represents the expected median increase for an intrauterine pregnancy. A percentile of 1 represents slowest expected increase for a normal pregnancy (ie, 99% of all pregnancies have this rate or faster). Percentile 99 represents the fastest expected increase (ie, only 1% have this rate or faster).

An additional important clinical caveat noted in the interpretation of these data is that, when a woman presents in this clinical situation where a gestation is not visualized with ultrasonography and the initial hCG value is less 5,000 mIU/mL, it is not necessary to know the date of her last menstrual period. The curve generated with our data is almost identical if the initial value of the curve is date of presentation or LMP. The reason for the slightly wider CI for the curve generated based on data from women in whom the LMP was known (Fig. 2) is likely due to a smaller sample size and some inherent error in their estimate of LMP. However, the data for the predicted hCG change for this subgroup is virtually identical to that derived from our entire data set. In other words, the curve predicting the rise in hCG should be the same for all women who have a last menstrual period 95 days or less from date of presentation (regardless of whether that date of LMP is known).

It is important to note that the data presented in this article reflect how hCG values should change over time for a woman with a viable intrauterine pregnancy. The data cannot be used to determine the viability of a pregnancy by extrapolating that a given hCG concentration is too low for a given gestational age.

The determination of the expected rise in hCG for a viable intrauterine pregnancy is the first step in the diagnosis of a nonviable pregnancy. It has been demonstrated repeatedly that the rate of rise of hCG in women with a miscarriage or ectopic pregnancy is slower than that of a viable pregnancy.⁴⁻¹³ If the rate of rise of serial hCG values is outside of the expected range for a normal gestation, the diagnosis of a nonviable gestation is made, but the location of the gestation has not been determined. It is important to understand that the converse of that statement is not true. Moreover, the finding of a “normal” rise in serial hCG values does not exclude the possibility of a miscarriage or ectopic pregnancy. Women with a miscarriage or an ectopic pregnancy can

have a rise that is within the range of a normal pregnancy. The ultimate determination of viability and location of a gestation must be made in conjunction with other diagnostic tests such as ultrasonography or evacuation of the uterus.

It is important that the commonly used “clinical rule” that a viable normal pregnancy will have an increase in the hCG concentration of 66% in 2 days⁴ was based on an 85% CI. Because of our large sample size, we are able to evaluate the tails of the distribution and base our estimates upon 99% CI. Our data demonstrate that at least a 53% increase in 2 days represents a minimal expected increase for a viable intrauterine pregnancy. The rate of rise can be considerably more rapid than these numbers.

Paramount to our patients who present with symptoms in the first trimester of pregnancy is the maintenance of a desired intrauterine pregnancy. This desire must be balanced with the need to reduce morbidity and mortality from a nonviable gestation such as an ectopic pregnancy. These new, more conservative, clinical rules derived from these data should allow a clinician to distinguish between a viable and nonviable pregnancy with a very low likelihood of prematurely interrupting a desired pregnancy.

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